

SuperZero: An AlphaZero-Style Agent for Ultimate Tic-Tac-Toe

Zamiul Rashid

Abstract—This report describes SuperZero, a program that learns to play Ultimate Tic-Tac-Toe. It follows the AlphaZero recipe: a neural network predicts which moves look good and who is winning, Monte Carlo tree search uses those predictions to look ahead, and the network is trained on the search’s own results. The network, MMNet, mirrors the game’s two levels (a 9×9 grid of cells and a 3×3 grid of boards); the training opponents are a mix of self-play, classical alpha-beta search and two external engines, and a C++ bitboard engine keeps the whole loop fast enough for a single workstation. We explain the game, the method, the implementation, and the measured results. The agent beats every classical baseline we built, including a depth-10 alpha-beta searcher with a 50-million-node cap (78.5% over 72 games), and the latest checkpoint (iteration 6000) finishes first in a seven-engine round robin, ahead of the strongest public engine we integrated, uttt.ai, with a 5–12–3 head-to-head record (55%).

I. INTRODUCTION

Ultimate Tic-Tac-Toe is Tic-Tac-Toe played on nine Tic-Tac-Toe boards at once, with one twist: the cell you pick decides which board your opponent must play in next. That twist links every local move to a global consequence, and it makes the game hard enough that exhaustive analysis only recently proved the first player’s forced win (43 moves at best, while the second player can hold out for at least 29 [1]).

The project’s goal was to build a strong player for this game, SuperZero, using the AlphaZero approach [2] on a laptop-class GPU rather than a data centre. This report explains what the game is, how AlphaZero-style learning works, how our implementation is put together, and what it achieved.

II. THE GAME

The board is a 3×3 grid of *local boards*, each a 3×3 Tic-Tac-Toe board, for 81 cells. X and O alternate, X first.

- A move marks an empty cell in an open local board.
- The *position* of that cell inside its board decides which board the opponent must play in next: play the top-right cell of any board and the opponent is sent to the top-right board.
- If that board is already closed (won or full), the opponent may play anywhere (a *free* move). The very first move is free.
- Three in a row wins a local board, which is then marked for that player on the *macro board*; a full board with no line is drawn. Nothing more is played in a closed board.
- Three local wins in a row on the macro board wins the game; all nine boards closed with no macro line is a draw.

Forced moves have at most 9 options, free moves up to 81. In code, a state is $s = (C, B, t, f, r)$: the 81 cell contents, the

9 board results, the player to move $t \in \{-1, +1\}$, the forced board $f \in \{-1, 0, \dots, 8\}$ (-1 = free), and the result r if the game is over. An action is one integer $a = 9b + c$ (board b , cell c), and every agent in the system—neural, alpha-beta, external—uses this same 81-action interface.

III. HOW ALPHAZERO-STYLE AGENTS WORK

AlphaGo, AlphaGo Zero and AlphaZero [2]–[4] share three ideas. Our system is a set of specific choices inside each.

a) *Tree search guided by a prior.*: Classical programs search the game tree with minimax and alpha-beta pruning [5], cutting branches that cannot change the answer and scoring the frontier with a hand-written heuristic. We keep such a searcher as a baseline, but the learning agent uses Monte Carlo tree search (MCTS), which grows a lopsided tree by repeating: *select* a path from the root, *expand* a leaf, *evaluate* it, *back up* the value. AlphaZero selects children with the PUCT rule [6]

$$a^* = \arg \max_a \left[Q(s, a) + c_{\text{puct}} P(s, a) \frac{\sqrt{N(s)}}{1 + N(s, a)} \right], \quad (1)$$

where Q is the average backed-up value of a child, N counts visits, and $P(s, a)$ is a learned *prior* that says which moves deserve attention. After a fixed number of *simulations*, the root visit counts $N(s, a)$ form an improved move distribution π , sharper than the prior.

b) *One network for policy and value.*: A network $f_\theta(s) = (\mathbf{p}, v)$ outputs the prior over moves and a scalar $v \in [-1, 1]$ estimating the result from the mover’s point of view. The prior concentrates a small search budget on plausible moves and the value replaces the horizon heuristic, which is why 512 guided simulations can compete with a brute-force search of tens of millions of nodes.

c) *Self-play policy iteration.*: The agent plays itself using search. Every position is stored with the search distribution π and the eventual outcome $z \in \{-1, 0, 1\}$, and the network is trained to imitate the search and predict the result, $\mathcal{L} = -\pi^\top \log \mathbf{p}_\theta + (v_\theta - z)^2$. A better network gives a better search, which gives better targets. AlphaZero starts from random weights on thousands of TPUs; on one GPU we had to bootstrap from a supervised dataset, mix in outside opponents, and add an extra training signal, as described next.

IV. OUR SYSTEM

A. Encoding and network

A position is encoded relative to the player to move as 289 floats: three one-hot cell planes (empty/mine/theirs, 81 each), four one-hot board planes (open/mine/theirs/drawn, 9 each)

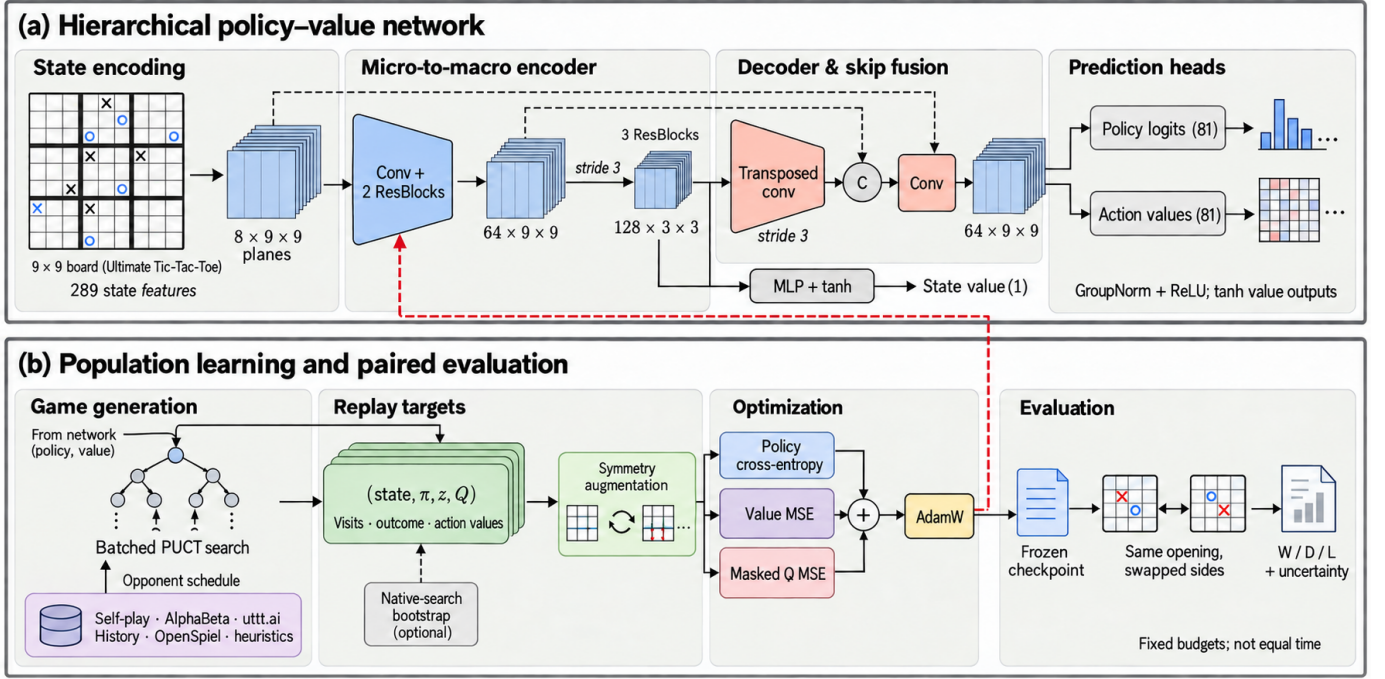


Fig. 1. SuperZero overview. (a) MMNet: eight 9×9 input planes $\rightarrow$ micro encoder $\rightarrow$ stride-3 pooling to a 3×3 macro grid $\rightarrow$ residual blocks $\rightarrow$ stride-3 upsampling $\rightarrow$ fusion with the micro features and input planes $\rightarrow$ policy and action-value heads; the state-value head reads the macro features. (b) Training: batched PUCT search against a scheduled opponent population produces (s, π, z, Q) rows that are symmetry-augmented and optimised with the three-term loss; frozen checkpoints are evaluated with mirrored seats.

and a 10-way one-hot for the forced board. Inside the network this vector is reshaped into eight 9×9 planes in true grid order, so neighbouring cells are neighbouring pixels.

The first network was a flat residual MLP (Linear(289, 512), three residual blocks) that treated the 81 cells as unrelated coordinates. It was replaced by MMNet (Table I, Fig. 1a), a hierarchical network modelled on the uttt.ai architecture [7]: convolutions on the 9×9 cell grid, a stride-3 convolution that pools each local board into one cell of a 3×3 macro grid, residual blocks on the macro grid, a stride-3 transposed convolution back to 9×9 , and U-Net-style skip connections [8]. The two resolutions are exactly the game’s two levels. The state value is read from the macro features (“who is winning” is a macro-board question); the policy and the per-move action values are read from the fused cell features. GroupNorm [9] replaces BatchNorm because one module serves both search batches (1–64 positions) and training batches (512–1024). MMNet has 1,560,835 parameters.

B. Search

The search is PUCT (Eq. 1) with $c_{\text{puct}} = 1.5$ and Dirichlet root noise ($\alpha = 0.3$, 25% mix), plus four additions that exist identically in the Python reference and the C++ engine:

- **Batched evaluation.** Up to L leaves (64 in training, 16 in evaluation) are collected and sent to the network in one call. While a leaf is pending, its path carries a *virtual loss* so that the next selection explores elsewhere.

TABLE I
MMNET CONFIGURATION (STTT/UNET.PY); SHAPES OMIT BATCH.

Stage	Operation	Output
Input	reshape/broadcast encoding	$8 \times 9 \times 9$
Stem	3×3 conv, GroupNorm, ReLU	$64 \times 9 \times 9$
Micro	$2 \times$ residual conv block	$64 \times 9 \times 9$
Pool	3×3 conv, stride 3	$128 \times 3 \times 3$
Macro	$3 \times$ residual conv block	$128 \times 3 \times 3$
Up	3×3 transposed conv, stride 3	$64 \times 9 \times 9$
Fuse	concat[micro, up, input]; 3×3 conv	$64 \times 9 \times 9$
Policy	1×1 conv	81 logits
Q head	1×1 conv, tanh	81 values
Value	flatten macro; MLP 1152 $\rightarrow$ 256 $\rightarrow$ 1, tanh	1

- **Proofs.** Terminal values are exact, a node whose children are all proven is itself proven, and proven values are never overwritten by network estimates. A proven root stops early.
- **Soft pruning.** After 8 visits, a child whose Q trails the best by more than 0.2 gets a reduced exploration weight (never below 0.2), and every 64th selection revisits the neglected child.
- **Subtree reuse.** The chosen child’s subtree becomes the next root.

After each search, the mean value $q_a = W(s, a)/N(s, a)$ of every visited root child is recorded along with a mask of which moves were visited.

C. Training targets and loss

Each stored position carries the encoding x , the visit distribution π , the legal mask, the game outcome z from the

TABLE II
TRAINING OPPONENTS, GAMES PER 100
(CONFIGS/POPULATION/BASELINE.JSON).

Family	Quota	Budget
Self-play	30	learner search, both sides
uttt.ai	25	64 / 128 / 256 simulations
AlphaBeta	25	depth 4–8; 100k–500k nodes
Historical checkpoints	10	128 / 256 / 512 simulations
Best checkpoint	5	128 / 256 / 512 simulations
Tactical / threat bots	3	depth 2
OpenSpiel MCTS	1	128–1024 rollouts / decision
Behavioural styles	1	fixed policies plus noise

mover’s view, and the root action values q with their visited mask m^Q . The loss is the AlphaZero objective plus a masked action-value regression taken from uttt.ai:

$$\mathcal{L} = - \underbrace{\sum_{a \text{ legal}} \pi_a \log p_\theta(a | s)}_{\text{policy}} + \underbrace{(v_\theta(s) - z)^2}_{\text{value}} + \underbrace{\frac{\sum_a m_a^Q (\hat{q}_\theta(s, a) - q_a)^2}{\sum_a m_a^Q}}_{\text{action value}}. \quad (2)$$

The third term exists because a game outcome is one bit per game and the policy target only says where the search looked; the Q term gives every visited move an explicit target, even in lost games. Training uses AdamW [10] (weight decay 10^{-4}), gradient clipping at 1.0, FP16 on CUDA, and eightfold symmetry augmentation applied consistently to planes, π , q and masks.

D. Bootstrap

Instead of starting from random weights, the C++ engine plays tens of thousands of games with a *model-free* heuristic MCTS and alpha–beta opponents, recording the same (s, π, z, q) rows, and the network is pretrained on them for ten epochs (cosine schedule $3 \times 10^{-4} \rightarrow 10^{-5}$ with a 100-step warm-up). The first bootstrap runs produced a value head that output exactly -1.0 for every position. The cause was not the architecture (the flat MLP died identically) but the learning rate: the head’s input is a large all-positive ReLU vector, so at $\eta = 10^{-3}$ the first two Adam steps push the pre-tanh activation to -5.27 , where the gradient is effectively zero, and it never recovers. A lower default (3×10^{-4}), warm-up, and an abort when the value output’s standard deviation drops below 0.02 fixed it.

E. Opponent population

Pure self-play can produce a player that is strong against itself and brittle against anything else, so each training game’s opponent is drawn from a quota-controlled population (Table II). Quotas are per 100 games and carry across iterations; the cursor is checkpointed. In self-play both sides’ positions are stored; in opponent games only the learner’s—the opponent shapes what the learner sees, but its moves are never used as policy labels.

Two external engines sit in the population. **uttt.ai** [7] is a published AlphaZero-like engine with a deployed ONNX network, trained upstream for about ten weeks; we vendor its Python reference at a pinned revision and run its own search (fresh root each move) in a persistent subprocess. **OpenSpiel** [11] supplies DeepMind’s official environment with rollout MCTS; its budget is per decision because it splits a free move into a board choice and a cell choice. A failing engine fails the game rather than being silently replaced.

F. Training loop

Each iteration plays 16 games at 512 simulations per move across eight worker processes, appends the rows to a 200,000-position replay buffer, runs a fixed number of AdamW steps on sampled minibatches, and periodically snapshots the network for the historical population. Checkpoints hold model, optimiser, replay, RNG, FP16 scaler and schedule state, so a run resumes exactly. Training ran in phases (Fig. 2): iterations 187–4193 on the original settings, a continuation to 4785 on a remote host, then—after the search fix of Sec. VI—a resumed run from 4786 with a cosine schedule from 2×10^{-4} to 10^{-5} , 16 workers from iteration 4948, and reanalysis switched on from 5138 to 6000. In total 83,552 training games were logged at a median of 18.8 s per iteration. The first flat-MLP runs had used a constant learning rate and oscillated once the replay buffer filled; all U-Net runs use cosine annealing.

The last phase added *reanalysis*: after each iteration, decision points from games the learner lost (prioritising opponents it is scoring below 50% against—in practice always the three uttt.ai settings) are re-searched from both sides with four times the budget (2,048 simulations), and the refreshed policy and Q targets are added to replay, capped at 25% of new positions [12]. Over iterations 5138–6000 this added 100,376 rows (about 116 per iteration, 19% of each sampled minibatch), flagged 2,860 suspected mistakes (4 proven), and cost about 6 s per iteration.

V. ENGINEERING

a) Bitboard engine.: Game operations dominate search cost, and the pure-Python `State` manages about 2×10^5 moves/s. The C++ engine (`cpp/`) packs a full position into 46 bytes: a 9-bit occupancy mask per player per board, three 9-bit macro masks, and one byte each for turn, forced board and result. A board’s (or the macro board’s) win status is one lookup into a 512-entry table indexed by the 9-bit mask, and legal moves are enumerated with bit-scan instructions in about 7 ns. The Python extension exposes a drop-in `FastState`, a batched encoder, an alpha–beta searcher and a `FastTreeSearch` with bit-identical visit counts to the Python search (Table III). Correctness rests on 2,000 random games (118,000 moves) played in lockstep between the Python and C++ states plus boundary tests; the suite has 533 tests.

b) Workers and one inference owner.: Self-play runs in eight spawned CPU processes, each owning its game and tree. None holds the network: leaf batches go to a single inference process that merges requests from all workers (up to 128

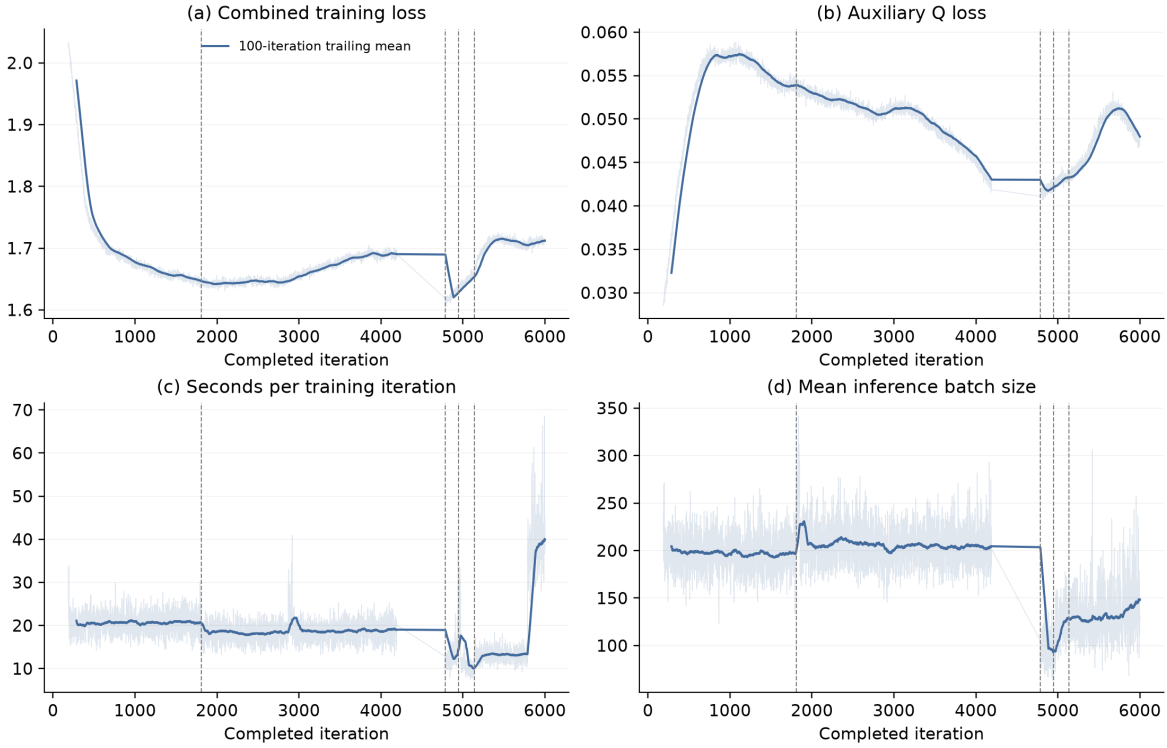


Fig. 2. Online training, iterations 187–6000 (raw and 100-iteration trailing mean). Dashed lines mark resumes at 1807, 4786 (post-fix), 4948 (16 workers) and 5138 (reanalysis on); iterations 4194–4785 were logged on another host and are bridged. The drop in mean inference batch after 4786 is the batched-search fix flushing partial batches. Loss is only a health signal: opponents and replay contents change over time.

TABLE III
PYTHON VS. C++ THROUGHPUT (CORE I7-12650H, SINGLE THREAD
UNLESS NOTED; G/S = GAMES/S). MULTIPLIERS ARE LOCAL TO THIS
MACHINE.

Operation	Python	C++	Speed-up
Legal-move gen.	338 k/s	145 M/s	430×
Move application	230 k/s	377 M/s	1640×
Encode to 289 floats	103 k/s	5.5 M/s	54×
Rollouts, 1 thread	5.8 k g/s	708 k g/s	123×
Rollouts, 10 threads	—	5.07 M g/s	882×
Alpha-beta, depth 3	67.9 ms	0.13 ms	520×
MCTS, NN callback	11.8 k sim/s	211 k sim/s	17.8×

positions), evaluates them on the GPU in FP16, and returns priors, values and action values. This keeps one copy of the weights in GPU memory and the GPU busy with large batches. The end-to-end self-play speed-up from the native search was only $1.71\times$: once game logic is fast, the network dominates. Every evaluation writes a manifest with checkpoint hashes, settings, library versions and the native build id, plus per-game rows.

VI. EVALUATION AND RESULTS

a) Method.: A checkpoint is evaluated in a round-robin championship. All entrants are copied and hashed first. Each matchup is played as *opening pairs*: a seeded random two-ply opening is played twice with the seats swapped, so first-move advantage cancels. A game scores $1/\frac{1}{2}/0$; a matchup’s score is $(W + \frac{1}{2}D)/(W + D + L)$. Uncertainty is a 95%

bootstrap interval over opening pairs (10,000 resamples), since the two games of a pair are not independent. Bayesian-Elo ratings anchored at AlphaBeta d4 = 1500 are *pool-local*, not universal. Baselines: C++ alpha-beta at depths 4, 6 and 10 (50-million-node cap), two shallow heuristic bots (*tactical*, *threat-block*), OpenSpiel MCTS at 100 rollouts per decision, and uttt.ai at 128 simulations. The candidate uses 512 simulations and leaf batch 16 (CPU inference for iterations 4193 and 4530, CUDA for 6000). Budgets are as configured, not equalised for time or compute.

b) Results.: Table IV reports three checkpoints. At iteration 4193 (seven entrants, 20 games per matchup) the agent wins 95–100% against every local baseline and scores 35% against uttt.ai (1–12–7). At iteration 4530 the roster grew to eight entrants with 72 games per matchup (2,016 games); the agent finishes second by score (85.8%, Elo 2201) behind uttt.ai (91.3%, Elo 2303) and ahead of depth-10 alpha-beta (71.2%, Elo 1960), scoring 41.0% against uttt.ai (9–41–22). At iteration 6000, trained with the fixed search and reanalysis, the agent finishes *first* in a seven-entrant, 420-game round robin (Fig. 3): 91.2% score and Elo 2342 against uttt.ai’s 90.4% and 2325, with a 5–12–3 head-to-head record (55%, CI 42.5–70%). Every other entrant lost all 20 games to both. The uttt.ai matchup is only ten opening pairs and its interval includes 50%: the two engines are now close, not separated. Against a depth-10 alpha-beta with a 5×10^7 -node cap, iteration 6000 scores 78.5% over 72 games (48–17–7, CI 70.1–86.8%), against 71.5% for iteration 4530 over the same number of games and 67.5% and

TABLE IV

CANDIDATE RESULTS AT THREE CHECKPOINTS (20, 72 AND 20 GAMES PER OPPONENT). SCORE COUNTS A DRAW AS HALF A WIN; BRACKETS ARE 95% OPENING-PAIR BOOTSTRAP INTERVALS. ITERATIONS 4193 AND 4530 PREDATE THE BATCHED-SEARCH FIX; 6000 WAS TRAINED AND EVALUATED AFTER IT. [‡]ALPHABETA D10 WAS A CHAMPIONSHIP ENTRANT AT ITERATION 4530 AND A SEPARATE 72-GAME MATCH SET AT ITERATION 6000, ON A DIFFERENT OPENING CORPUS.

Opponent	Iteration 4193		Iteration 4530		Iteration 6000	
	W-D-L	Score (%)	W-D-L	Score (%)	W-D-L	Score (%)
AlphaBeta d10 [‡]	—	—	46–11–15	71.5 [63.9, 79.2]	48–17–7	78.5 [70.1, 86.8]
AlphaBeta d6	19–0–1	95.0 [85, 100]	63–5–4	91.0 [85.4, 95.8]	19–0–1	95.0 [85, 100]
AlphaBeta d4	18–2–0	95.0 [87.5, 100]	70–1–1	97.9 [94.4, 100]	19–1–0	97.5 [92.5, 100]
Tactical	19–0–1	95.0 [85, 100]	71–1–0	99.3 [97.9, 100]	20–0–0	100
Threat-block	20–0–0	100	72–0–0	100	20–0–0	100
OpenSpiel MCTS	20–0–0	100	72–0–0	100	20–0–0	100
uttt.ai (128)	1–12–7	35.0 [25, 45]	9–41–22	41.0 [35.4, 46.5]	5–12–3	55.0 [42.5, 70]
All games	97–14–9	86.7	403–59–42	85.8	103–13–4	91.2
Pool rank (Elo)	2nd (2138)		2nd (2201)		1st (2342)	

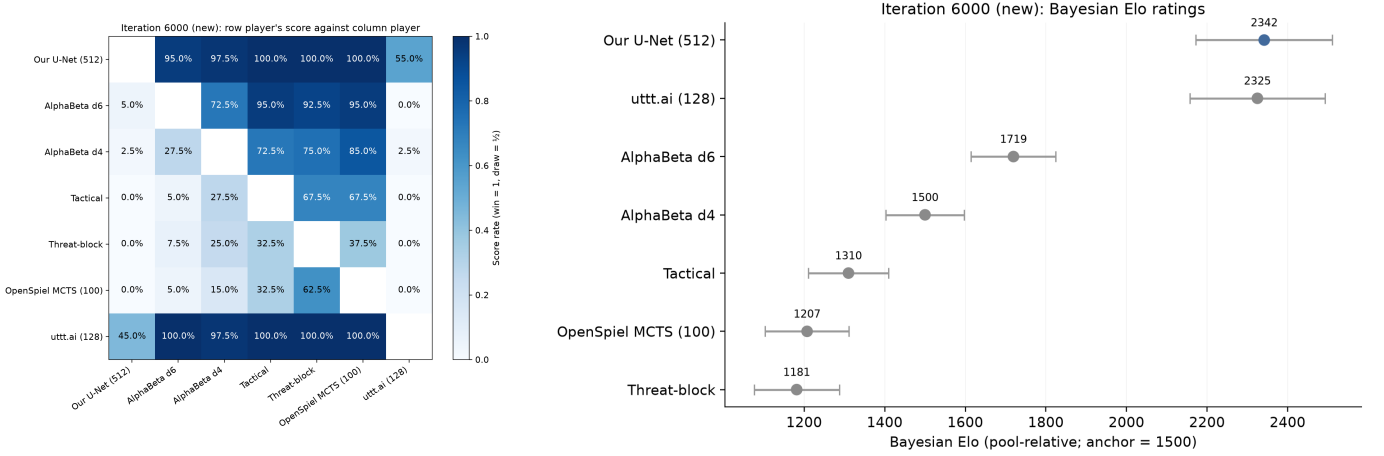


Fig. 3. Iteration-6000 championship. Left: round-robin score matrix (row player’s score against column player). Right: pool-local Bayesian-Elo ratings with 95% CIs, anchored at AlphaBeta d4 = 1500.

65% for iterations 1806 and 4193 over 20-game sets. The two 72-game intervals overlap and use different opening corpora, so this is a consistent trend, not a measured improvement. A 20-game check of the 4530 checkpoint at 2,000 simulations against uttt.ai scored 40%, far too small a sample to say anything about search budget.

c) *Search audit.*: Because the gap to uttt.ai was stable, we audited the search with stub evaluators and known-outcome positions on both backends. Value perspective, terminal handling, legal moves and Python/C++ parity all passed. One bug was found: in batched search, once all unexplored branches were pending, selection kept returning the same terminal leaf, each return counted against the budget, and the search could exhaust its simulations before any pending evaluation was applied; with leaf batch 64 in self-play, endgame targets could be distorted. The fix flushes the batch after any terminal backup while evaluations are pending; its footprint is the drop in mean inference batch at iteration 4786 in Fig. 2. The 4193 and 4530 results predate the fix; the 6000 checkpoint was trained from 4786 onward and evaluated with it (native build 63a72ea). The audit also showed that virtual loss flattens root visit distributions at large leaf batches (top-move share 0.84 at batch 1 vs. 0.06 at batch 64 on the empty board), so

self-play trains the policy on softer targets than a sequential search would give.

VII. DISCUSSION

The pipeline works: an agent trained on one GPU beats a 50-million-node depth-10 alpha-beta searcher 78.5% of the time, every shallower baseline above 90%, and now holds its own against uttt.ai. The move from the flat MLP to MMNet coincided with the first checkpoints that draw rather than lose against uttt.ai (in the flat-MLP era uttt.ai went 137–2–1 against the whole field), and the head-to-head score has climbed 27.5% → 35% → 41% → 55% across iterations 1806, 4193, 4530 and 6000.

What closed the gap between 4530 and 6000 cannot be attributed: the search fix, reanalysis, 1,470 more iterations and the decaying learning rate all changed together, and the 6000 head-to-head is ten opening pairs. The remaining differences from uttt.ai are also open: its network is about three times larger and was trained on roughly 8 million searched positions per stage; our policy targets at leaf batch 64 are flattened by virtual loss; and we regress the value to the game outcome where uttt.ai uses the search value. Each is one controlled experiment away—a larger uttt.ai match set, a reanalysis on/off

comparison, leaf-batch and value-target ablations—and the launchers exist.

Two limitations apply to every number here. Budgets are configured simulation counts, not equal wall-clock or compute (the 6000 candidate also used CUDA inference, and the 4530 championship ran alongside a training job). And uttt.ai, OpenSpiel and alpha-beta all appear in the training population—uttt.ai is the very opponent reanalysis was prioritising—so these scores do not measure transfer to unseen opponents.

VIII. CONCLUSION

SuperZero is a complete AlphaZero-style system for Ultimate Tic-Tac-Toe: exact rules in Python and C++, a two-level policy/value/action-value network, batched tree search with proofs, a bootstrapped and population-trained learning loop, and a paired evaluation harness. It dominates every classical baseline we could construct and, at iteration 6000, finishes level with the strongest public engine we integrated. The two most useful findings came from audits rather than training—a learning-rate-driven value-head collapse and a batched-search starvation bug—and both are now covered by tests. The next step is controlled experiments—above all a larger uttt.ai match set—not more features.

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